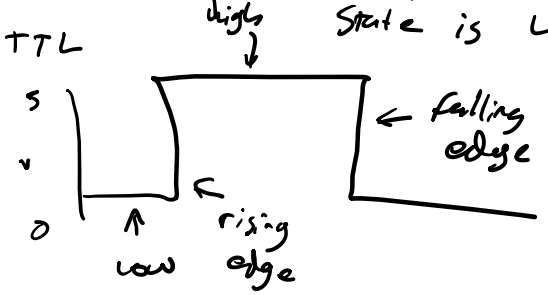
Conceptually → what is sisdig?

- Signal & Logic

ex. CNE focused on electronic amplification and gain manipulation.

Meanwhile in SD, Signal is only 0 or 1.
Voltage is 0V or 5V.
State is LOW or HIGH.



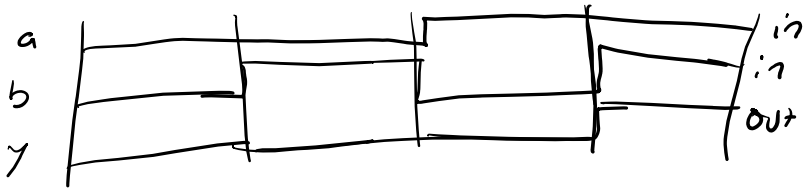
No longer are we concerned about components now it's about logic & gates.



one of the representations is a Timing Diagram.

Shows what happens internally signal-wise when processing input-output.

ex.



this more represents states of I/O in a logical context.

(rangkaian kombinasional)

From here, it stems into a truth table.

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Circuit manipulating data so that the

Sisdig: Manipulating data so that the output is what we want.

all gates are represented as mathematical formulas.

and = ·
or = +
not = - / ~

Boolean algebra:
a set of rules to simplify a given logic without changing its functionality.

ex:
I will bring an umbrella if it rains And i walk.
A B
"result" (Y) "variables" "logic clause"

Y = A and B → A · B

Boolean Rules

AND Rule
A · 0 = 0
A · 1 = A
A · A = A
A · $\bar{A}$ = 0

OR RULE
A + 1 = 1
A + 0 = A
A + A = A
A + $\bar{A}$ = 1

NOT RULE
$\bar{\bar{A}}$ = A

(a+b)(a+c) = a+bc

a + $\bar{a}$b = a+b

a + a·b = a

distributive
a(b+c) = ab+ac
a+(bc) = (a+b)(a+c)

commutative
a·(b·c) = (a·b)·c
a+(b+c) = (a+b)+c

demorgan
$\overline{(a+b)}$ = $\bar{a}\cdot\bar{b}$
$\bar{a}\cdot\bar{b}$ = $\overline{a+b}$

When to use boolean? max 3 var.

ex: $ab\bar{c} + a\bar{b}\bar{c} + b\bar{c}$
= $a\bar{c}(b+\bar{b}) + b\bar{c}$
= $a\bar{c} + b\bar{c}$

ex = $ab + ab\bar{c} + a\bar{b}\bar{c}$

SIMPLIFY IN UP

$= ac + bc$

ex: $ab + a\bar{b}c + a\bar{b}\bar{c}$

$= a(b + \bar{b}c + \bar{b}\bar{c})$

$= a(b + c + b\bar{c})$

$= a(b + c + \bar{c})$

$= a(b + 1)$

$= a$

ex: $(a+b+c)(a+\bar{b}+c)(a+b+\bar{c})$

assume $x = (a+b)$

$(x+c)(a+\bar{b}+c)(x+\bar{c})$

$= (x + c\cdot\bar{c})(a+\bar{b}+c)$

$= (x+0)(a+\bar{b}+c)$

$= (a+b)(a+\bar{b}+c)$

$= a + b(\bar{b}+c)$

$= a + b\cdot\bar{b} + bc$

$= a + bc$

Minterm notation:

$\Sigma(a,b,c) = \Sigma(3,5,6,7)$ - SOP

maxterm notation:

$Y(a,b,c) = \Pi(0,1,2,4)$ - POS

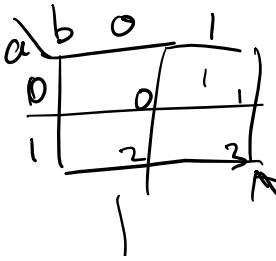
bool alg. doesn't show most optimized results, so enter:

K-Map.

rules:

- find the truth table
- map it into a matrix

a	b	Y
0	0	0
0	1	1
1	0	0
1	1	0



← once drawn, then simplify.

indexing.

requires = 2^n

boxes = 4, 2, 1



requires = 2^n
blocks = 4, 2, 1
gather the 1's
(horiz & vert)

$y = \bar{a}b$